EXPERIMENT – 2 **DATE: 22-10-2026**

FORMAL DESIGN OF AN EMBEDDED CONTROLLER USING STATE MACHINE (STATECHART) MODELING – TRAFFIC LIGHT CONTROLLER

Objectives of this Experiment:

1. To understand the formal design of finite state machines (FSM / Statechart modeling) for embedded real-time controllers.
2. To interface traffic signaling LEDs (Red, Yellow, Green) and pedestrian/mode push buttons with the Arduino UNO platform.
3. To configure and read digital input states from push buttons using pull-down resistor network architectures.
4. To implement time-driven and event-driven state transitions using Embedded C programming.
5. To control output devices (LED indicators) according to active finite state chart conditions.
6. To transmit continuous real-time state, timing, and system status logs to the Proteus Virtual Terminal via UART Serial Communication.
7. To simulate, test, and formally verify statechart execution and timing precision within the Proteus schematic capture environment

OUTPUT PARAMETER TABLE – VALUES TO BE RECORDED

S. No.	Output Parameter	Expected Value / Condition	Practical / Simulated Observation
1	Initial State	S0 – NS Green / EW Red	Verified
2	State S1	NS Yellow / EW Red	Verified
3	State S2	NS Red / EW Green	Verified
4	State S3	NS Red / EW Yellow	Verified
5	S0 Duration	10 seconds	Verified
6	S1 Duration	3 seconds	Verified
7	S2 Duration	10 seconds	Verified
8	S3 Duration	3 seconds	Verified
9	State Sequence	S0 → S1 → S2 → S3 → S0	Verified
10	Virtual Terminal	Correct state messages displayed	Verified
11	Continuous Operation	State sequence repeats continuously	Verified

RUBRIC:

Objective	Assessment Criterion	Weightage	Marks
Develop the Proteus circuit	Circuit Design and Configuration	2.5	
Develop the Embedded C program	Program Development	2.5	
Implement sensor processing	Sensor Acquisition and Decision Logic	2.5	
Verify system operation	Simulation and Result Analysis	2.5	
Total		**10**	

Experiment uploaded in GitHub on :

https://github.com/madurivadapalli-tech/192311368.EMBEDDED/blob/main/LAB%20EXPERIMENTS/EXPERIMENT%202.11368.pdf